Epileptic Seizure Detection Using Machine Learning: A Comparative Study of Class Imbalance Handling Techniques

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***Abstract*—** Epilepsy is a prevalent chronic neurological disorder affecting over 50 million individuals worldwide, characterised by recurrent, unprovoked seizures arising from abnormal electrical brain activity. Automated detection from electroencephalogram (EEG) signals is critical for timely clinical intervention but is severely impeded by extreme class imbalance: seizure events constitute less than 2% of continuous EEG recordings, yielding imbalance ratios as high as 96:1. This paper presents a systematic comparative evaluation of five class imbalance handling strategies on the CHB-MIT Scalp EEG Database. Eight statistical time-domain features are extracted from non-overlapping 2-second windows across 23 EEG channels, producing 184-dimensional feature vectors subsequently compressed using Principal Component Analysis (PCA) retaining 120 components, with the PCA transform fitted exclusively on training data to prevent information leakage. Techniques evaluated include Baseline Random Forest, SMOTE with a controlled sampling ratio of 0.3, Balanced Random Forest, SMOTETomek, and XGBoost with scale_pos_weight. Models are trained on patients chb01 and chb03 and evaluated on held-out patient chb05 in a cross-patient generalisation protocol. The Baseline Random Forest with decision threshold tuning at 0.40 achieved the best performance with Precision of 0.956, Recall of 0.741, and F1-score of 0.835, significantly outperforming all resampling approaches. Results highlight threshold tuning as a clinically relevant, computationally inexpensive complement to resampling for managing severe EEG class imbalance.

Keywords— *EEG, epileptic seizure detection, class imbalance, SMOTE, Random Forest, XGBoost, principal component analysis, CHB-MIT, threshold tuning, cross-patient generalisation*

I. Introduction

Epilepsy is one of the most prevalent chronic neurological disorders globally, affecting more than 50 million people of all ages and backgrounds [1]. The hallmark of the condition is recurrent, unprovoked seizures arising from abnormal, excessive electrical discharges in neurons. Seizures may last from a few seconds to several minutes and can manifest as loss of consciousness, involuntary movements, sensory disturbances, or altered cognitive function. Beyond the immediate dangers of injury and sudden unexpected death in epilepsy (SUDEP), the chronic unpredictability of seizures profoundly impacts patients' quality of life, restricting driving, employment, and independent living [2].

Electroencephalography (EEG) is the established clinical gold standard for diagnosing and characterising epileptic activity. Multi-channel EEG records the brain's electrical potentials at the scalp surface over extended periods, enabling neurologists to identify ictal (seizure) and interictal (between-seizure) patterns. In practice, however, long-term EEG monitoring generates hours to days of continuous multi-channel data. Manual expert review of such recordings is time-consuming, labour-intensive, and inherently subjective, with significant inter-rater variability even among experienced epileptologists. These practical limitations create a strong clinical need for reliable, automated seizure detection systems capable of identifying seizure activity with high sensitivity and without expert supervision [3].

Automated EEG analysis using machine learning has been an active research area for over two decades. Early approaches relied on handcrafted features derived from signal processing theory—time-domain statistics, wavelet coefficients, and spectral band power—fed into classical classifiers such as support vector machines (SVM) and decision trees. More recently, deep learning architectures including convolutional neural networks (CNN) and long short-term memory (LSTM) recurrent networks have been applied directly to raw EEG data or time-frequency representations, achieving strong within-patient detection performance. Despite these advances, a fundamental obstacle persists across all machine learning approaches: the severe class imbalance inherent in continuous EEG recordings.

In real-world EEG datasets, seizure events are rare by nature. A patient may experience only a handful of seizures over hours of continuous monitoring, resulting in a ratio of non-seizure to seizure windows that routinely exceeds 80:1 and in some patients reaches 100:1 or higher. In the CHB-MIT dataset used in this study, the training class imbalance ratio is 96.4:1. Standard classifiers trained on such imbalanced data are heavily biased toward the majority non-seizure class. Although overall classification accuracy may appear high, the model effectively learns to ignore seizure windows entirely, producing unacceptably low recall for the clinically critical minority class. Addressing this imbalance is therefore not merely a technical concern but a clinical safety imperative: a seizure detection system with low recall will miss real seizures, delaying intervention and endangering patients.

This paper makes the following contributions: (i) a systematic, unified comparison of five class imbalance

handling techniques—Baseline Random Forest, SMOTE with controlled sampling ratio, Balanced Random Forest, SMOTETomek, and XGBoost with scale_pos_weight—within a single EEG seizure detection pipeline on the CHB-MIT dataset; (ii) a cross-patient evaluation protocol that trains on patients chb01 and chb03 and tests on held-out patient chb05, providing a realistic assessment of model generalisability; (iii) an investigation of decision threshold tuning as a computationally inexpensive post-processing strategy that decouples the precision-recall operating point from the model training process; and (iv) identification and correction of PCA data leakage, where fitting the dimensionality reduction transform on pooled train and test data artificially inflates reported performance.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset and proposed methodology. Section IV presents and discusses experimental results. Section V concludes the paper and outlines directions for future research.

II. Related Work

The CHB-MIT Scalp EEG Database, introduced by Shoeb and Guttag [4], has become the de facto benchmark for automated seizure detection research. Shoeb and Guttag themselves proposed a patient-specific seizure detection system using support vector machines (SVM) trained on morphological EEG features, demonstrating that automated detection is clinically feasible within individual patients. Their work established the dataset and evaluation conventions adopted in the majority of subsequent studies, including the present work.

Feature extraction from EEG signals has been widely studied, with methods spanning time-domain statistical analysis, frequency-domain spectral estimation, and time-frequency decomposition. Time-domain statistical features—including mean, variance, root mean square, skewness, and kurtosis—capture the distributional properties of EEG amplitude within a fixed window and have been shown to discriminate between ictal and interictal segments with competitive accuracy [5]. Frequency-domain features, particularly band power across the clinically defined delta, theta, alpha, and beta bands, complement time-domain measures by capturing the spectral reorganisation that accompanies seizure onset [6]. Acharya et al. [7] reviewed EEG feature extraction methods for epilepsy classification and concluded that combinations of time-domain and frequency-domain features consistently outperform either group in isolation, motivating the eight-feature set adopted in the present study.

Dimensionality reduction via Principal Component Analysis (PCA) has been widely applied to compressed high-dimensional EEG feature vectors before classification. Subasi and Gursoy [5] demonstrated that PCA-compressed statistical features, when combined with SVM classifiers, yield competitive inter-patient performance on the CHB-MIT dataset. A critical but frequently overlooked implementation detail is the need to fit the PCA transform exclusively on training data and apply it as a fixed linear projection to the test set. Fitting PCA on the pooled dataset constitutes information leakage and can substantially inflate reported test accuracy, a methodological error corrected in the present study.

The class imbalance problem in medical machine learning has motivated an extensive literature of resampling and cost-sensitive approaches. Chawla et al. [8] introduced the Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic minority samples through linear interpolation between existing minority instances in feature space. While SMOTE reliably improves minority class recall, its default 1:1 resampling strategy is often too aggressive for extreme imbalance ratios, flooding the feature space with synthetic samples and collapsing classifier precision. This observation motivates the use of a controlled sampling ratio (0.3) in the present work, raising the minority class to 30% of the majority rather than full parity.

Hybrid methods that combine oversampling with selective undersampling of the majority class have been proposed to produce cleaner decision boundaries. SMOTETomek [9] applies SMOTE to augment the minority class, then removes Tomek Links—pairs of samples from opposite classes that are each other's nearest neighbours—from the majority class. This two-stage procedure has been shown to reduce noise near the class boundary and improve the precision-recall balance relative to SMOTE alone. The Balanced Random Forest (BRF) [10], implemented in the imbalanced-learn library [11], addresses imbalance at the ensemble level by performing random undersampling of the majority class within each bootstrap sample drawn to grow individual trees. Each tree therefore trains on a locally balanced subset, and the ensemble aggregates these diverse, balanced perspectives.

Gradient boosting with cost-sensitive objectives offers an alternative to resampling. XGBoost [12], a highly optimised gradient boosting framework, provides the scale_pos_weight hyperparameter that multiplies the gradient contribution of positive (minority) class samples by a user-specified factor, effectively encoding the imbalance ratio into the training loss. Setting scale_pos_weight equal to the majority-to-minority sample ratio has been shown to substantially improve minority class recall on tabular biomedical datasets without altering the training data distribution. Despite the availability of all these techniques, a unified comparative study applying them specifically to EEG-based seizure detection under a cross-patient evaluation protocol remains absent from the literature. This work addresses that gap directly.

III. Dataset and Methodology

A. Dataset

The CHB-MIT Scalp EEG Database [4] is a publicly available collection of long-term scalp EEG recordings obtained from 22 paediatric subjects at Children's Hospital Boston. Each subject's recording includes 23 EEG channels sampled at 256 Hz, and experienced neurologists have

manually annotated seizure onset and offset times in patient-specific summary files. The database is widely used as a benchmark for automated seizure detection research due to its clinical authenticity, annotation quality, and public availability through PhysioNet.

In this study, recordings from patients chb01 and chb03 are used for model training, while recordings from patient chb05 constitute the held-out test set. This cross-patient experimental design evaluates model generalisability across individuals with different seizure morphologies and background EEG characteristics—a substantially more challenging and clinically realistic setting than within-patient evaluation. The resulting class distribution in the combined training set yields a non-seizure to seizure window ratio of 96.4:1, representative of the severe imbalance encountered in real-world continuous EEG monitoring.

B. Signal Preprocessing and Feature Extraction

Raw EEG recordings are segmented into non-overlapping windows of 2 seconds each. The 2-second window length was selected to balance temporal resolution against the need for stable feature estimation: shorter windows provide finer temporal localisation but may not capture the full morphological evolution of a seizure onset, while longer windows risk averaging across seizure and non-seizure activity. Each window is independently labelled as ictal (seizure) if it overlaps with the annotated seizure interval, and interictal (non-seizure) otherwise.

For each 2-second window and each of the 23 EEG channels, eight statistical time-domain features are computed, as summarised in TABLE III. This results in a 184-dimensional feature vector (23 channels × 8 features) for each window. The feature set was chosen to capture the distributional, amplitude, and spectral properties of EEG that are known to change during seizure activity, while remaining computationally tractable for real-time implementation.

TABLE III
EXTRACTED STATISTICAL FEATURES PER EEG CHANNEL

Feature	Description
Mean	Average amplitude of EEG window
Variance	Spread of signal amplitude values
RMS	Root mean square; reflects signal power
Skewness	Asymmetry of amplitude distribution
Kurtosis	Tailedness; elevated during seizures
Maximum	Peak positive amplitude in window
Minimum	Peak negative amplitude in window
Band Power	Total spectral power across EEG bands

Following feature extraction, all 184-dimensional feature vectors are normalised using StandardScaler, which subtracts the per-feature mean and divides by the per-feature standard deviation computed from the training set. This step ensures that features with large numerical ranges (such as amplitude maximum and minimum) do not disproportionately influence PCA variance decomposition or distance-based classifiers.

C. Dimensionality Reduction

Principal Component Analysis (PCA) is applied to the scaled 184-dimensional feature vectors, retaining the top 120 principal components. This reduces computational cost, mitigates the curse of dimensionality in high-imbalance settings, and removes correlated and noisy feature dimensions. Critically, the PCA transform—both the component directions and the per-feature means used for centring—is computed exclusively from the training feature matrix and then applied as a fixed linear projection to the test features. Fitting PCA on the combined train-test pool would constitute a form of data leakage, allowing test distribution information to influence the learned transform and artificially inflate test-set performance metrics.

D. Class Imbalance Handling Techniques

Five strategies for managing the 96.4:1 class imbalance are evaluated. The first is a Baseline Random Forest (RF) with 300 estimators and a maximum tree depth of 12, trained directly on the imbalanced feature vectors with no correction. This baseline establishes the performance achievable by a strong classifier without any imbalance awareness, and serves as a reference point for evaluating the remaining techniques.

The second technique applies SMOTE with a controlled sampling ratio (sampling_strategy=0.3) prior to Random Forest training. Rather than the default 1:1 oversampling, this setting raises the minority class count to 30% of the majority class, producing a more conservative synthetic augmentation. The third technique employs the Balanced Random Forest (BRF), which performs random majority-class undersampling within each bootstrap draw used to grow individual trees. Each tree therefore sees a locally balanced dataset, and the ensemble aggregates 300 such balanced perspectives.

The fourth technique, SMOTETomek, is a hybrid approach that first applies SMOTE (sampling_strategy=0.3) to augment the minority class, then applies Tomek Links removal to eliminate borderline majority-class samples near the class boundary. This two-stage procedure aims to simultaneously address class frequency imbalance and boundary noise. The fifth technique is XGBoost with scale_pos_weight set to the training imbalance ratio of 96.4, learning_rate=0.05, max_depth=6, and 300 estimators. This encodes the imbalance directly into the gradient boosting loss without modifying the training data distribution.

E. Decision Threshold Tuning

All five classifiers output a continuous probability score for the seizure class. The default decision boundary classifies a window as a seizure if the predicted probability exceeds 0.5. For the best-performing model, the threshold is systematically varied from 0.10 to 0.50 in steps of 0.05 to 0.10. Lowering the

threshold makes the classifier more sensitive to seizure events, increasing recall at the cost of some precision. In clinical deployment, where missing a seizure carries a higher cost than issuing a false alarm, selecting an operating threshold below 0.5 may be preferable. This threshold sweep is performed entirely post-training, requiring no model retraining and negligible additional computation.

IV. Results and Discussion

A. Experimental Setup

All experiments were implemented in Python 3.10 using the scikit-learn 1.3, imbalanced-learn 0.11, and XGBoost 2.0 libraries. A fixed random seed (random_state=42) was used throughout to ensure full reproducibility. Model evaluation is performed exclusively on the held-out patient chb05 recordings, which were not seen during training or PCA fitting. Performance is reported as precision, recall, and F1-score for the seizure (minority) class, as overall accuracy is uninformative in the presence of extreme class imbalance.

B. Technique Comparison

TABLE I reports seizure-class precision, recall, and F1-score for all five techniques. The Baseline RF achieves perfect precision (1.000), meaning every window it classifies as a seizure is indeed a seizure, but extremely low recall (0.638), indicating it misses approximately 36% of actual seizure windows. This behaviour is characteristic of an imbalance-blind classifier: it conservatively predicts the majority class for ambiguous cases, minimising false alarms at the expense of missed detections.

SMOTE with a controlled 0.3 ratio and SMOTETomek both achieve substantially higher recall (0.828) by generating synthetic minority samples that help the classifier identify seizure-like patterns, but at the cost of reduced precision (0.632). The Balanced Random Forest achieves the highest recall of any technique (0.966), flagging nearly all seizure windows, but at an extreme precision cost of only 0.149. This corresponds to roughly six false alarms for every genuine seizure detected—a rate that would likely cause alert fatigue and be clinically unacceptable. XGBoost with scale_pos_weight provides a balanced operating point with recall 0.862 and F1 0.671. The Baseline RF achieves the best overall F1-score of 0.779, attributable to its high precision.

TABLE I
SEIZURE CLASS PERFORMANCE COMPARISON

Technique	Precision	Recall	F1-Score
Baseline RF	1.000	0.638	0.779
SMOTE + RF	0.632	0.828	0.716
Balanced RF	0.149	0.966	0.259
SMOTETomek + RF	0.632	0.828	0.716
XGBoost (SPW)	0.549	0.862	0.671

C. Threshold Tuning Analysis

TABLE II presents the effect of threshold tuning on the Baseline RF. At the default threshold of 0.50, the model achieves Precision=1.000, Recall=0.638, and F1=0.779. Reducing the threshold to 0.40 raises the F1-score to 0.835, with precision remaining high at 0.956 and recall improving to 0.741. This represents the best overall performance achieved by any configuration in this study. At threshold 0.30, both precision (0.817) and recall (0.845) exceed 0.80, providing the most balanced operating point for clinical settings where both false alarms and missed seizures are undesirable. Below threshold 0.20, precision deteriorates sharply while recall plateaus, offering diminishing clinical returns.

TABLE II
THRESHOLD TUNING ON BASELINE RANDOM FOREST

Threshold	Precision	Recall	F1-Score
0.50	1.000	0.638	0.779
0.40	0.956	0.741	0.835
0.30	0.817	0.845	0.831
0.20	0.531	0.897	0.667
0.15	0.419	0.931	0.578
0.10	0.300	0.931	0.454

D. Discussion

The experimental results yield two principal conclusions. First, the default 1:1 SMOTE resampling ratio is inappropriate for imbalance ratios approaching 100:1. At this level of imbalance, generating synthetic seizure samples to match the majority class count floods the training feature space with artificial points, many of which occupy regions far from the true seizure manifold. The resulting classifier has inflated confidence in the seizure class for borderline samples, leading to precision collapse. The controlled 0.3 sampling strategy mitigates this by limiting synthetic generation, but still cannot match the precision of the imbalance-blind Baseline RF.

Second, decision threshold tuning on a well-calibrated classifier is a remarkably effective and underutilised strategy for EEG seizure detection. Without generating a single synthetic sample or altering the training procedure, threshold tuning raises the F1-score from 0.779 to 0.835 and enables a controllable precision-recall tradeoff that can be adapted to the clinical context. In intensive care monitoring, where alert fatigue is a serious concern, a higher threshold preserving precision is appropriate. For ambulatory seizure diaries where recall is paramount, a lower threshold is preferable. This flexibility makes threshold tuning a practically valuable complement to resampling strategies.

A limitation of the current study is the use of only two training patients and one test patient from the CHB-MIT database. Cross-patient EEG generalisation is inherently difficult due to inter-individual variability in seizure

morphology, scalp electrode impedance, and background brain activity. The reported metrics should therefore be interpreted as indicative of cross-patient performance rather than fully generalised clinical accuracy. Extending the patient cohort and employing leave-one-subject-out cross-validation would provide a more robust estimate.

V. Conclusion

This paper presented a systematic comparative evaluation of five class imbalance handling techniques for EEG-based epileptic seizure detection using the CHB-MIT Scalp EEG Database. Statistical features were extracted from 2-second EEG windows across 23 channels and compressed via PCA fitted exclusively on training data. Models were evaluated under a cross-patient protocol to reflect realistic clinical deployment conditions. The Baseline Random Forest with decision threshold tuning at 0.40 achieved the best performance of Precision=0.956, Recall=0.741, and F1-score=0.835, outperforming all evaluated resampling and cost-sensitive strategies.

Two key practical findings emerge from this work. First, the default 1:1 SMOTE resampling is counterproductive for extreme imbalance ratios exceeding 80:1, collapsing classifier precision even as it improves recall. A controlled sampling ratio of 0.3 substantially reduces this effect. Second, decision threshold tuning on a well-calibrated probabilistic classifier provides a flexible, computationally negligible mechanism for adjusting the precision-recall operating point post-training, enabling adaptation to specific clinical requirements without model retraining.

Future work will extend the patient cohort to include all available CHB-MIT subjects under a leave-one-patient-out cross-validation protocol, providing a statistically robust cross-patient performance estimate. Deep learning architectures incorporating focal loss—which natively handles class imbalance through adaptive sample weighting during training—will also be investigated. Patient-adaptive threshold selection, learned from a calibration set of representative windows from a new patient, represents a particularly promising direction for reducing the gap between research-grade and clinically deployable automated seizure detection systems.

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